HDOC-69

Recursion - Fibonacci Series and Recursive Call Count

Question

Write a recursive function to generate the first n Fibonacci numbers. Also count the number of recursive calls made.

Algorithm / Logic

1. Input the value of n, the number of Fibonacci terms to generate.

2. Use a recursive function fibonacci(n) with base cases fibonacci(0) = 0 and fibonacci(1) = 1.

3. For n > 1, return fibonacci(n - 1) + fibonacci(n - 2).

4. Increment a global counter every time the recursive function is entered. This counts all calls, including base-case calls.

5. Call fibonacci(i) for i = 0 to n - 1 and print each returned value.

6. After generating the series, display the total number of recursive calls made.

C Program

```
#include <stdio.h>

long long callCount = 0;

long long fibonacci(int n)
{
   callCount++;

   if (n == 0)
      return 0;
   if (n == 1)
      return 1;

   return fibonacci(n - 1) + fibonacci(n - 2);
}

int main()
{
   int n, i;

   printf("Enter the number of Fibonacci terms: ");
   scanf("%d", &n);
```

```
    if (n <= 0)
    {
        printf("Please enter a positive integer.\n");
        return 0;
    }

    printf("First %d Fibonacci numbers are:\n", n);
    for (i = 0; i < n; i++)
        printf("%lld ", fibonacci(i));

    printf("\nTotal number of recursive calls = %lld\n", callCount);

    return 0;
}
```

Explanation

The function fibonacci() calculates a Fibonacci term recursively. The first two terms are handled directly by the base cases. Every other term is obtained by adding the two preceding Fibonacci terms.

The global variable callCount is incremented whenever fibonacci() is called. Since the program invokes fibonacci() separately for every term from 0 to n - 1, the final counter represents the total number of recursive function invocations required to print the complete series.

This direct recursive method is easy to understand, but it repeats many calculations. Therefore, its running time grows rapidly as n increases.

Test Cases

Test Case	Input n	Fibonacci Output	Recursive Calls
1	1	0	1
2	5	0 1 1 2 3	19
3	7	0 1 1 2 3 5 8	53
4	0	Invalid input message	0

Sample Output

```
Enter the number of Fibonacci terms: 7
First 7 Fibonacci numbers are:
0 1 1 2 3 5 8
Total number of recursive calls = 53
```

Result

Thus, the program successfully generates the first n Fibonacci numbers using recursion and counts the total number of recursive function calls made while generating the series.